DASE7506 Mini-project 2 Report

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Abstract

In this project, I study whether bounded mathematical and question-alignment evidence can improve selection among four language-model solutions. The proposed Groupwise Relative Evidence Correction (GREC) method groups numerically equivalent answers, freezes a linear backbone using the course's fixed representations, and learns a regularized correction from 70 condition, local-checking and cross-candidate features. Relative to the same backbone, it gains 24 and 40 questions in two training-only three-fold partitions and improves validation selection from 746/798 to 751/798. Its frozen test record is 401/467 (85.87%), using 22,399 bytes of inference assets and approximately 3.05 seconds for fresh prediction plus independent scoring on the local CPU. Matched ablations support relative evidence, while failure cases show that numerical consistency does not establish correct interpretation.

1 Task and research question

The course task supplies the question, four fixed candidate solutions and 1024-dimensional candidate representations [1]. The selector returns one index. Any candidate with the reference's final numerical value is accepted, including equivalent fractions and decimals. This is *final-answer selection*, not proof grading or new answer generation. The bank derives from GSM8K [2]; traces and representations were prepared by the course using Qwen3-0.6B. I use no additional training data or pretrained weights.

My question is: *can local mathematical evidence, especially cross-candidate comparisons, add information beyond a representation-based answer-group scorer?* I expect comparisons to help when a majority repeats an error or a trace returns an intermediate quantity. However, rules can miss semantic errors or penalize a correct-final-answer trace with flawed reasoning. I therefore learn a soft correction rather than reject every flagged candidate. This report presents the basic development path and key comparisons.

2 Method

2.1 Official parsing and answer groups

Let a_{ij} denote the value returned by the unchanged official parser for candidate j of question i. Unparseable candidates are excluded from final selection. Candidates with equal values form $G_{ia} = \{j : a_{ij} = a\}$, and $\mathcal{A}_i$ contains distinct valid answers. For candidate scores z_{ij}, define

$$g_{ia} = |G_{ia}|^{-1} \sum_{j \in G_{ia}} z_{ij}. \tag{1}$$

I select the group with maximum score and return its earliest candidate; ties follow candidate order. If every answer on a new input is unparseable, index zero is returned. The implementation does not depend on public-bank IDs or a fixed number of questions.

2.2 Joint linear backbone

The input $x_{ij} \in \mathbb{R}^{1070}$ concatenates 1024 fixed coordinates, 22 earlier text/numerical features and 24 consensus interactions. The 22 include length, formatting, question-word/number coverage, answer validity and numerical properties. The consensus block contains support, log-count, support gap and interactions between support or answer diversity and seven numerical features. This backbone retains surface features; it is not the separate no-style backbone.

Training-only means and scales standardize x, with a 10^{-6} scale floor. Let $b_{ij} = w^\top \bar{x}_{ij}$ and pool these scores using Eq. (1). The objective is

$$\mathcal{L}_{\text{base}} = \frac{1}{N} \sum_i \left[\log \sum_{a \in \mathcal{A}_i} \exp(g_{ia}) - g_{i,a_i^\star} \right] + \frac{0.01}{2} \|w\|_2^2, \tag{2}$$

where $a_i^\star$ is the correct numerical answer. Multiple correct candidates map to its one answer group. Full-batch CPU Adam uses learning rate 0.03, zero initialization and 360 epochs. A common intercept cancels within a question and is not trained.

2.3 Three evidence blocks

The reference-free vector $\phi_{ij} = [S_{ij}, V_{ij}, R_{ij}] \in \mathbb{R}^{70}$ contains the following blocks.

Conditions and target (S, 28 dimensions). Features track question numbers used in operations, salient multiplicities, restricted target-unit agreement and computed results not used later. Additional signals combine total, remainder, profit or splitting cues with candidate quantities and recognize limited explicit multiplicative relations. They are alignment heuristics, not a complete entity or temporal parser.

Local verification (V, 22 dimensions). A bounded AST interpreter uses rational arithmetic for supported numerical expressions, limited LaTeX fractions, known unit conversions and polynomial identities up to degree two. It distinguishes checked, inconsistent and unknown expressions, and compares explicit conclusions with the official parsed answer. Unsupported expressions, mixed-scale operations and cross-currency conversions are skipped. Symbolic nonidentities are soft signals, not universal proofs of error. Text length, expression complexity and magnitudes are bounded; candidate code is never executed.

A restricted repair hypothesis replaces an inconsistent intermediate claim by its evaluated value and propagates it through later supported expressions. Its result only supports or contradicts an existing candidate; it does not generate another answer.

Relative evidence (R, 20 dimensions). Features compare consistency, condition coverage, units and conclusions across candidates: quality ranks, Pareto comparisons, same-answer quality, support interacting with detected errors, answer ratios matching question factors, and support from another trace's conclusion or repair hypothesis. Since R is derived partly from S/V information, removing the *direct* S or V input does not eliminate that information everywhere.

2.4 Frozen-backbone residual

I freeze w and train only $u \in \mathbb{R}^{70}$:

$$z_{ij} = b_{ij} + u^\top \widetilde{\phi}_{ij}, \qquad \widetilde{\phi}_{ij} = \text{clip}\left(\frac{\phi_{ij} - \mu_\phi}{\max(\sigma_\phi, 0.05)}, -8, 8 \right). \tag{3}$$

Training uses the same group likelihood as Eq. (2), with penalty $0.03\|u\|_2^2/2$. L-BFGS uses at most 100 iterations, 140 evaluations, history size 12 and strong-Wolfe line search. The full-data residual fit takes 28 iterations. The retained model has 1070 backbone and 70 residual coefficients; only the latter are optimized in stage two. The fixed offset directly tests whether evidence improves an existing strong scorer.

This model, denoted GREC, has no extra vote reward ($\alpha = 0$), although consensus already enters its backbone. It is a learned evidence residual, not majority voting followed by a hard mathematical filter.

3 Protocol and rationale for model GREC

3.1 Data and evaluation

The splits contain 4755 training, 798 validation and 467 test questions, with four candidates each. Training references supervise fitting; validation supports development. Outputs are graded by the unchanged independent scorer. Validation has 443 questions with one distinct valid answer and 355 with competing answers, whereas every test question has at least two distinct valid answers. Thus validation and test accuracy reflect different difficulty mixes.

There are 2114 training questions with competing valid answers; a single-answer group has zero answer-group discrimination loss. The main evidence study uses two training-only three-fold partitions, with seeds 211 and 509, stratified by correct-candidate count, answer diversity and valid-candidate count. Each question is held out once per partition. Backbones, preprocessing and residuals are fitted within each training portion. The two OOF predictions of the same question are dependent, they are not 9510 independent examples.

3.2 Development evidence supporting GREC

My scientific focus is a *single joint answer-group backbone plus an interpretable evidence correction*. Within that family, the methodological rationale is to preserve the strong scorer, require paired OOF improvement, compare blocks and regularization under a matched setup, and seek corroboration on validation. Keeping the backbone fixed makes the contribution of the new evidence easier to study than changing the entire scoring architecture or introducing another ensemble-selection layer.

Method	OOF_{211}/4755	OOF_{509}/4755	Val./798
Joint backbone	4364 (91.78%)	4353 (91.55%)	746 (93.48%)
Frozen residual, S only	4378 (92.07%)	4375 (92.01%)	748 (93.73%)
Frozen residual, V only	4380 (92.11%)	4364 (91.78%)	745 (93.36%)
Frozen residual, S+V	4382 (92.16%)	4373 (91.97%)	747 (93.61%)
Frozen residual, all; L2=.10	4382 (92.16%)	4385 (92.22%)	749 (93.86%)
Frozen residual, all; L2=.03 (GREC)	4388 (92.28%)	4393 (92.39%)	751 (94.11%)

Table 1: Development evidence

Among the eight originally evaluated joint frozen-residual configurations (four evidence sets and two L2 values), GREC has the highest mean OOF count. Compared with its own backbone, it gains 24 and 40 OOF questions, with positive changes in all six folds (Table 2). Validation increases from 746 to 751: eight errors are fixed and three correct choices are broken. Within the complete residual family, L2=0.03 exceeds 0.10 in both OOF partitions and validation.

Partition seed	Fold 1 gain	Fold 2 gain	Fold 3 gain	Total gain
211	4	17	3	24
509	18	21	1	40

Table 2: GREC's paired OOF gains over its own joint backbone. Seeds describe data partitions, not random initializations of a neural network.

4 Results and key comparisons

4.1 Basic development path

Table 3 retains the steps relevant to GREC. Accuracy tables report correct counts followed by percentages in parentheses; the paired-fold table reports gains in question counts rather than accuracy. Vector scoring improves on majority vote; masking unparseable answers prevents choices that necessarily receive no credit. Hard majority-first selection is weaker than allowing the vector score to override consensus. A candidate-group objective then aligns training with choosing any correct candidate. The joint answer-group backbone and evidence residual extend this path.

Method	Validation /798 (Acc.)	Test /467 (Acc.)
Majority vote	688 (86.22%)	323 (69.16%)
Independent vector BCE	716 (89.72%)	358 (76.66%)
Vector BCE + valid-answer mask	720 (90.23%)	361 (77.30%)
Hard vote-first + vector tie-break	708 (88.72%)	353 (75.59%)
Vector BCE + soft vote	723 (90.60%)	369 (79.01%)
Vector candidate-group objective + soft vote	734 (91.98%)	371 (79.44%)
Joint answer-group backbone	746 (93.48%)	391 (83.73%)
Evidence residual (GREC)	751 (94.11%)	401 (85.87%)

Table 3: Selected milestones, spanning several development rounds. They are not all single-factor ablations. Test columns describe frozen versions rather than a test-ranking selection rule.

GREC reaches 751/798 (94.11%) on validation and records 401/467 (85.87%) on the frozen test bank, versus 746/798 and 391/467 for its backbone. On the 355 validation disagreement questions, counts are 308 and 303. The gain therefore occurs on competitive questions, rather than the single-answer subset.

4.2 Matched direct-input ablations

I fix the joint backbone, folds, preprocessing procedure, group objective, residual optimizer, L2=0.03 and $\alpha = 0$. The original study includes no residual, S, V, $S + V$ and $S + V + R$. I subsequently complete the subsets with R, $S + R$ and $V + R$: 21 residual fits across six folds and full-data training, reusing the matching fold-local backbones. A no-backbone control reuses the evidence-only fit with $\alpha = 0$.

Direct residual inputs	OOF_{211}/4755	OOF_{509}/4755	Val./798	Test/467
none	4364 (91.78%)	4353 (91.55%)	746 (93.48%)	391 (83.73%)
S	4378 (92.07%)	4375 (92.01%)	748 (93.73%)	394 (84.37%)
V	4380 (92.11%)	4364 (91.78%)	745 (93.36%)	392 (83.94%)
R	4382 (92.16%)	4382 (92.16%)	749 (93.86%)	396 (84.80%)
S+V	4382 (92.16%)	4373 (91.97%)	747 (93.61%)	394 (84.37%)
S+R	4384 (92.20%)	4389 (92.30%)	750 (93.98%)	398 (85.22%)
V+R	4390 (92.32%)	4379 (92.09%)	748 (93.73%)	399 (85.44%)
S+V+R (GREC)	4388 (92.28%)	4393 (92.39%)	751 (94.11%)	401 (85.87%)
S+V+R, no backbone	4099 (86.20%)	4100 (86.23%)	690 (86.47%)	328 (70.24%)

Table 4: All direct evidence-input subsets. S: conditions; V: local checks; R: relative evidence. Missing subsets and the matched no-backbone evaluation were supplemented post hoc, not used as new selection data.

Removing R loses 6 and 20 OOF questions, four validation questions and seven fixed-test questions. Removing direct V loses four OOF questions in each partition and one validation question. Removing direct S is less stable: it gains two OOF questions in the first partition but loses 14 in the second and three on validation. Consequently, I do not claim every block helps every split. Evidence without a

backbone gives only 690 validation and 328 test questions, indicating that rules cannot replace the learned representation.

These are *direct-input* ablations: R retains information derived from S/V, and the backbone retains earlier numerical/text features. They do not identify three independent semantic abilities. Removed training columns and their fitted weights are exactly zero. The unchanged predictor still extracts all blocks. Its timing therefore does not measure a separately engineered feature-pruned implementation.

4.3 Regularization, refitting and fusion

L2=0.10 gives 749 validation and 397 test questions for the full residual, below GREC's corresponding records. Refitting the full backbone and evidence with L2=0.03 gives 751 validation and 396 test questions, but changes both the optimizer and the trainable parameter set. It is not a clean single-factor test of freezing, and I cannot attribute its difference solely to keeping the backbone fixed.

Method	Mean OOF/4755	Val./798	Test/467
Joint backbone	4358.5 (91.66%)	746 (93.48%)	391 (83.73%)
GREC	4390.5 (92.33%)	751 (94.11%)	401 (85.87%)
Enhanced joint + enhanced no-style, 50/50	4395 (92.43%)	750 (93.98%)	397 (85.01%)
Enhanced joint + enhanced no-style, 75/25	4394 (92.41%)	750 (93.98%)	399 (85.44%)
Three-member blend	4392 (92.37%)	749 (93.86%)	395 (84.58%)

Table 5: Representative fixed combinations. Mixture scores are normalized by their training within-question standard deviation and use fixed small vote rewards; GREC has no additional vote term.

The enhanced joint/no-style 75/25 pair records 399 test questions, and the three-member blend 395, compared with GREC's 401.

5 Errors and reliability

5.1 Representative paired validation cases

GREC fixes eight backbone errors and introduces three, leaving 47 validation errors. The examples below use zero-based validation positions for traceability, not as inference keys.

Arithmetic repair (65). A five-day sequence is $3, 5, 7, 9, 11$. The old choice writes the sum as 36; GREC selects the existing candidate giving 35. This is a local inconsistency for which the correct alternative is already available.

An omitted component (473). Three skirts require 144 square feet and the bodice another 12, at \$3 per square foot. The old trace computes the bodice but omits it from the final total, returning 432. GREC selects $(144 + 12) \times 3 = 468$. Valid arithmetic is insufficient if the wrong quantities are combined.

A semantic regression (626). For a 4×6 rectangle covered by a 4×4 square, the old choice correctly gives $24 - 16 = 8$. GREC selects a trace that invents a third dimension and returns $96 - 64 = 32$. The arithmetic is internally consistent but answers a volume question instead of an area question. Restricted unit and target heuristics do not prevent this error.

Correct value with flawed reasoning (338). The correct change after buying water is $20 - (100 \times 3/50) \times 2.5 = 5$. GREC selects a trace ending in 5, but the trace incorrectly says the buyer has \$10 and discusses $10 - 15 = -5$ before giving the positive answer. The scorer accepts it despite unsound reasoning. Thus increased final-answer accuracy is not evidence that every selected explanation is valid.

These cases show evidence associated with changes; they do not prove which feature caused a switch. Successes alone would hide the semantic regressions and coincidentally correct final values.

5.2 Coverage and uncertainty

At least one supported local step is found in 2791/3192 validation candidates (87.44%); 361 have a numerical-inconsistency flag. Only 130 have a checked unit conversion and 43 a unit-inconsistency flag. Coverage is not detector precision: the labels concern final values, not individual reasoning steps.

For the 355 disagreement questions, the margin is the highest answer-group score minus the second highest. Sorting the fixed model's margins gives Table 6. Accuracy increases across quartiles, suggesting useful uncertainty ordering on this development sample. No calibration map or rejection threshold is fitted. Margins are not probabilities, and reused validation data do not establish independent calibration performance.

Answer-group margin quartile	Correct / questions	Accuracy (%)
1 (lowest)	62/89	69.66
2 (middle)	74/89	83.15
3 (middle)	85/89	95.51
4 (highest)	87/88	98.86

Table 6: Descriptive GREC reliability on validation disagreement questions. Sorted-margin quartiles contain 89/89/89/88 questions.

6 Resources and reproduction

Experiments use Windows 11, an Intel Core i9-12900H and four CPU threads. The observed environment is Python 3.12.14, NumPy 2.5.3 and PyTorch 2.7.1+cu126; training tensors are on CPU and no pretrained weights are loaded. Prediction requires only NumPy and standard-library modules.

GREC's configuration and uncompressed NPZ total 22,399 bytes, including weights and preprocessing statistics. Fresh full-test prediction plus an independent scoring process takes 3.047 seconds locally, including startup, imports, loading, feature extraction and scoring. This single run is not a robust microbenchmark. The 16 MiB and 30-second limits ultimately apply on the reference Xeon Platinum 8457C, which has not been measured here.

A full-data joint-backbone fit takes about 3.54 seconds and GREC's residual fit about 0.26 seconds, excluding feature preparation and the wider search. The main evidence round takes 471.14 seconds through training/freezing: 41 primary configurations across six folds and full-data fits, 21 shared backbone fits and 13 combinations. Two extra vote-scale controls were evaluated separately. These training/freezing times exclude the independent validation/test prediction and scoring runs for all variants. The supplemental round takes 8.68 seconds through freezing, using cached features and backbones and adding 21 residual fits. Earlier exploration incurred additional costs recorded per round in the repository. These timings are not an end-to-end total for all historical experimentation and are not the cost of training only GREC.

From `code/`, set `OMP_NUM_THREADS`, `OPENBLAS_NUM_THREADS` and `MKL_NUM_THREADS` to 4:

```
python -X utf8 -B predict.py --method evidence_models.py --checkpoint ../checkpoints/grec --split test --output
    runs/GREC-test-predictions.json
python -X utf8 -B score_predictions.py --split test --predictions runs/GREC-test-predictions.json --output runs/
    GREC-test-score.json
```

The exact model identifier is `joint_residual_relative_r3`. Its inference import closure contains eight local modules, listed by the repository audit; copying only the entry point is insufficient. Asset and source hashes are in `code/runs/evidence_round/freeze.json`.

Existing checks cover Torch/NumPy export agreement, empty and variable-size inputs, unknown IDs, reordering, unparseable candidates, bounded expression handling and original-file integrity. Saved GREC test predictions re-score consistently.

7 Conclusion

In this project, I developed GREC, which combines a joint answer-group linear backbone with a regularized correction learned from 70 condition, local-verification and relative-evidence features. Numerically equivalent answers are grouped and scored together, and the model selects one of the four supplied solutions without generating a new answer.

GREC correctly selects 751 of 798 validation answers (94.11%) and 401 of 467 test answers (85.87%), improving on the joint backbone by 5 validation questions and 10 test questions. The two training-only OOF partitions improve by 24 and 40 questions, with gains in all six folds. The frozen model uses 22,399 bytes of inference assets and completes full-test prediction plus independent scoring in approximately 3.05 seconds on the local four-thread CPU.

AI assistance and reuse

OpenAI Codex provided assistance with implementation, experiment design, checks and analysis. The task, data, parser and scorer are course-provided; GREC reuses earlier components implemented within this project. Public-method explorations which are not needed to explain GREC are not individually listed here, the records and attribution remain in the repository. No additional pretrained weights or training datasets enter GREC, and frozen evaluation makes no LLM/API calls. Data licenses and Qwen preparation attribution are retained.

References

[1] DASE7506. *Mini-project 2: Selecting Reliable LLM Reasoning*. Course guide and fixed-bank protocol `dase7506-mp2-open-hard-verifier-v5`, 2026. Local guide: `guide/GUIDE.md`. `https://github.com/xudongwu-0/DASE7506-MP2`.

[2] OpenAI. *GSM8K: Grade School Math*, 2021. `https://github.com/openai/grade-school-math`. The course bank is a filtered and partitioned derivative; its candidate selection accuracy is not the original generation benchmark score. Attribution and license: `code/data/NOTICE.txt` and `code/data/LICENSE.txt`.